

Between Microbes and Codes: How Microorganisms Can Improve the World We Live In



Discover more about me and my work!



Dr. Anderson Santos de Freitas
Supervised by Dr. Siu Mui Tsai



My colleagues and I obtain microbial data from...



Soil



Trees



Functional Foods



Humans



Animals

...and convert it into...

Real World Solutions!



Main tools and skills



Main outcomes from my work



About our experts

Anderson Santos de Freitas has both a Bachelor's Degree in Biotechnology and a Master's in Biological Sciences earned at the Federal University of Pampa. He is working with Plant-Soil Feedback in degraded areas of the Central Amazon while earning his PhD at the Center for Nuclear Energy in Agriculture.

Dark earth: scientists unveil new 'secret weapon' to combat deforestation and climate change

With the increasing urgency for ecological restoration across the planet, a dark secret in the heart of the Amazon could provide a solution.

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By Noa Leach
Published May 5, 2023 at 1:15 am

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Rare Amazon dark soils could help forest restoration, study shows

Juliana Ennes
14 Jun 2023 Read



Current Research



AMAZONIAN DARK EARTH AND ITS MICROBIOTA AS A TOOL FOR THE RESTORATION OF DEGRADED SOILS

Grant number: 25/18420-3

Grouped support types: Scholarships in Brazil - Post-Doctoral

Duration: April 01, 2026 - March 31, 2029

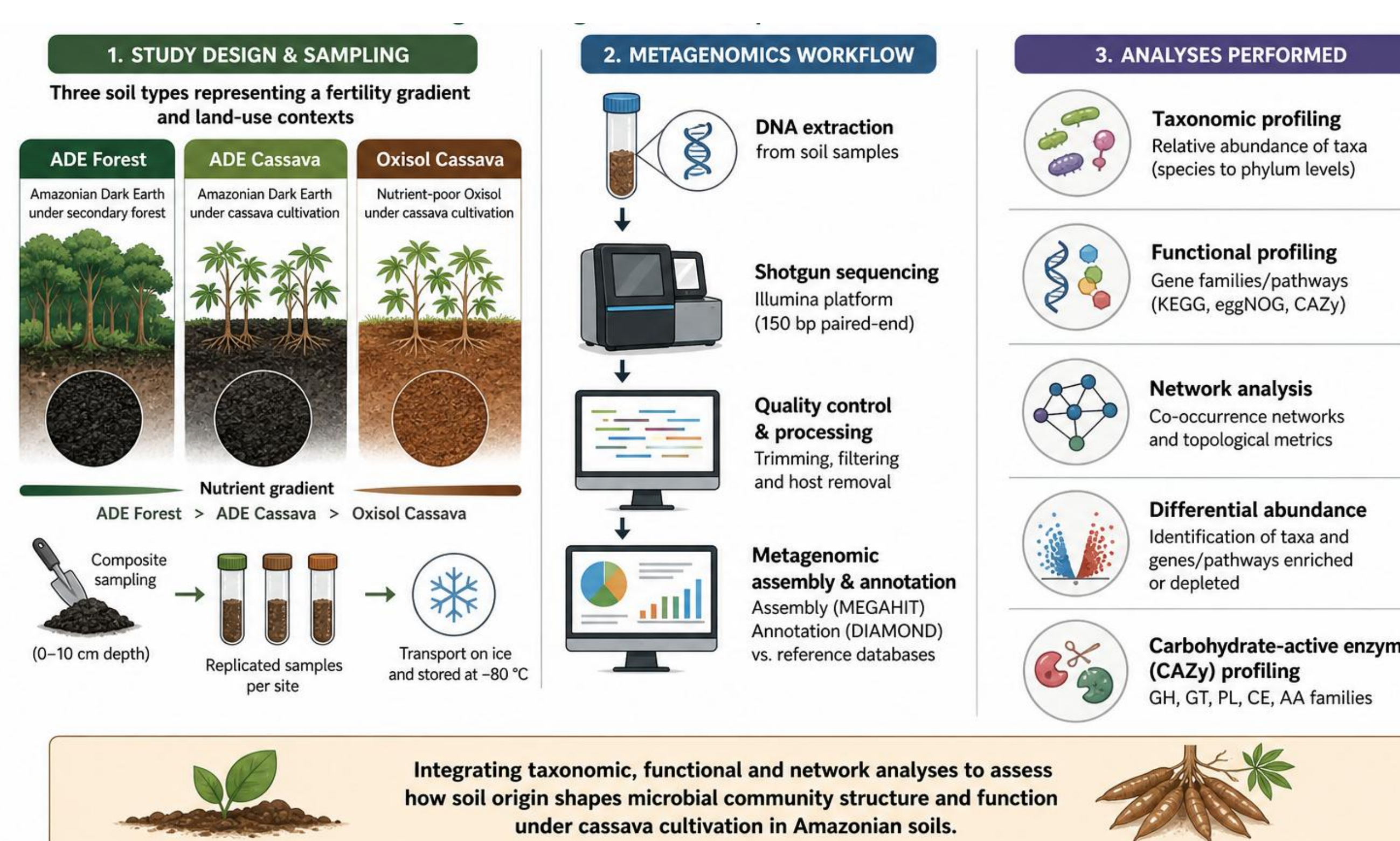
Field of knowledge: Agronomical Sciences - Agronomy - Soil Science

Principal Investigator: Tsai Siu Mui

Grantee: Anderson Santos de Freitas

Host institution: Centro de Energia Nuclear na Agricultura (CENA), Universidade de São Paulo (USP), Piracicaba, SP, Brazil

Associated research grant: 21/10573-4 - Center for Carbon Research in Tropical Agriculture (CCARBON), AP/CEPID



230+ isolates from ADE

Novel Genomes

N-fixers, ACC deaminase+, Siderophores

Genomic insights of *Fictibacillus terranigra* sp. nov., a versatile metabolic bacterium from Amazonian Dark Earths

Soil and Agricultural Microbiology – Research Paper | Published: 15 February 2024

Volume 55, pages 1817–1828, (2024) Cite this article

Thierry Alexandre Pellegrinetti, Izadora de Cássia Mesquita da Cunha, Miriam Gonçalves de Chaves, Anderson Santos de Freitas, Gabriel Schimmelpfeng Passos, Ana Vitória Reina da Silva, Simone Raposo Cotta, Siu Mui Tsai & Lucas William Mendes

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Boosting tree growth in the Amazon rainforest using Amazonian Dark Earths

Research | Open access | Published: 30 January 2026

Volume 26, article number 15, (2026) Cite this article

Anderson Santos de Freitas, Luís Felipe Guandalin Zagatto, Gabriel Silvestre Rocha, Franciele Muchalak, Guilherme Lucio Martins, Solange dos Santos Silva-Zagatto, Rogério Eiji Hanada, Aleksander Westphal Muniz & Siu Mui Tsai

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Harnessing the synergy of *Urochloa brizantha* and Amazonian Dark Earth microbiomes for enhanced pasture recovery

Research | Open access | Published: 17 January 2025

Volume 25, article number 27, (2025) Cite this article

Anderson Santos de Freitas, Luís Felipe Guandalin Zagatto, Gabriel Silvestre Rocha, Franciele Muchalak, Guilherme Lucio Martins, Solange dos Santos Silva-Zagatto, Rogério Eiji Hanada, Aleksander Westphal Muniz & Siu Mui Tsai

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ORIGINAL RESEARCH article

Front. Soil Sci. 04 May 2025
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Amazonian dark earths enhance the establishment of tree species in forest ecological restoration

Anderson Santos de Freitas^{1*}, Luís Felipe Guandalin Zagatto^{1†}, Gabriel Silvestre Rocha¹, Franciele Muchalak¹, Solange dos Santos Silva¹, Rogério Eiji Hanada¹, Aleksander Westphal Muniz¹, Siu Mui Tsai¹

JOURNAL ARTICLE
Land-use intensification reshapes microbial phosphorus cycling, organic matter composition, and phosphorus fractions in Amazonian soils
Guilherme Lucio Martins, Gabriel Gustavo Tavares Nunes Monteiro, Markus Lange, Anderson Santos de Freitas, Luiza do Nascimento Silva Barbosa, Johannes van Leeuwen, Jorge Emílio de Carvalho Soares, Rogério Eiji Hanada, Gerd Glöwen, Siu Mui Tsai

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